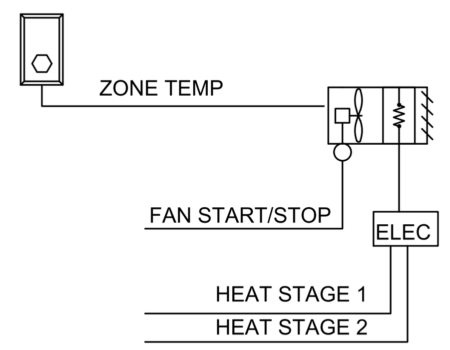


RUN CONDITIONS:
UNIT HEATER CONTROLS SHALL BE AN EXTENSION OF THE JCI CONTROL SYSTEM.
THE UNIT SHALL RUN TO MAINTAIN A SPACE TEMPERATURE SETPOINT OF 45°F (ADJ.).

FAN:
THE FAN SHALL RUN ANYTIME THE ZONE TEMPERATURE DROPS BELOW HEATING SETPOINT, UNLESS SHUTDOWN ON SAFETIES.

ELECTRIC HEATING STAGES:
THE CONTROLLER SHALL MEASURE THE ZONE TEMPERATURE AND STAGE THE HEATING TO MAINTAIN ITS HEATING SETPOINT. TO PREVENT SHORT CYCLING, THERE SHALL BE A USER DEFINABLE (ADJ.) DELAY BETWEEN STAGES, AND EACH STAGE SHALL HAVE A USER DEFINABLE (ADJ.) MINIMUM RUNTIME.

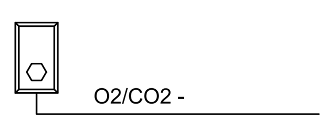
BAS:

PROVE FAN STATUS

PROVIDE FAN STATUS MISMATCH ALARM

UNIT HEATER STATE ALARM

ELECTRIC UNIT HEATER CONTROL SEQUENCE

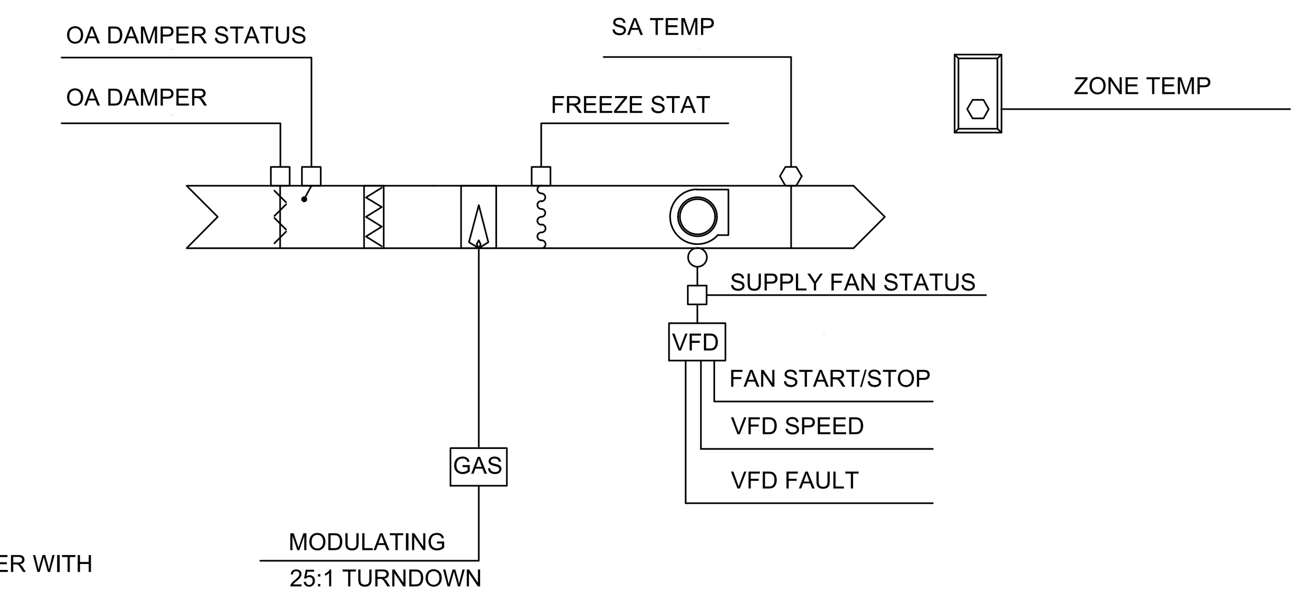


CARBON DIOXIDE (CO2) CONCENTRATION MONITORING:
THE CONTROLLER SHALL MEASURE THE ZONE CO2 CONCENTRATION.

ALARMS SHALL BE PROVIDED AS FOLLOWS:

- HIGH ZONE CARBON DIOXIDE CONCENTRATION: IF THE ZONE CO2 CONCENTRATION IS GREATER THAN 900PPM (ADJ.) WHEN IN THE OCCUPIED MODE.

TUNNEL CARBON DIOXIDE (CO2) CONCENTRATION MONITORING



MAKE UP AIR UNIT CONTROL SEQUENCE

THE UNIT SHALL BE PROVIDED FROM THE FACTORY WITH:

24VAC TRANSFORMER

FACTORY MOUNTED AND WIRED OUTDOOR AIR INLET DAMPER WITH ACTUATOR

MICROPROCESSOR CONTROLLER

THE MICROPROCESSOR CONTROL SHALL BE FACTORY PROGRAMMED, MOUNTED, WIRED AND TESTED. CONTROLLER SHALL HAVE A

LIGHTED LCD DISPLAY AND KEYPAD FOR CHANGING SET POINTS AND MONITORING UNIT OPERATION. THE CONTROLLER SHALL BE EQUIPPED WITH THE FOLLOWING SENSORS:

OUTDOOR AIR TEMPERATURE SENSOR

SUPPLY DISCHARGE TEMPERATURE SENSOR

BUILDING MANAGEMENT SYSTEM (BMS) COMMUNICATION

THE MICROPROCESSOR CONTROLLER SHALL BE CAPABLE OF INTEGRATING INTO A BUILDING MANAGEMENT SYSTEM (BMS) TO ALLOW THE BMS TO REMOTELY ADJUST SET POINTS, VIEW UNIT STATUS POINTS AND ALARMS. THE MICROPROCESSOR SHALL INCLUDE THE ABILITY TO COMMUNICATE OVER BACnet MSTP

SUPPLY FAN SEQUENCE

THE UNIT HAS BEEN PROVIDED WITH A FACTORY MOUNTED VARIABLE FREQUENCY DRIVE (VFD). THE VARIABLE FREQUENCY DRIVESHALL CONTROL THE SUPPLY FAN SPEED AS INDICATED BY THE FOLLOWING SEQUENCE:

CONSTANT VOLUME:

TUNNEL ENCLOSURE DAMPERS SHALL BE COMMANDED OPEN, PROVE OPEN STATUS VIA FEEDBACK BEFORE MAU IS ENABLED.

SUPPLY FAN SHALL BE LOCKED-OUT UNTIL ANY 2 (ADJ.) OF THE 6 TUNNEL ENCLOSURE DAMPERS HAVE PROVEN OPEN VIA FEEDBACK SIGNAL.

THE VFD SHALL BE PROGRAMMED FROM THE FACTORY FOR A CONSTANT SUPPLY FAN SPEED. THIS IS TO BE ADJUSTED FOR AIR BALANCING ONLY AND IS NOT TO BE MODULATED.

IF MAU FAILS, LOCKOUT AND CLOSE TUNNEL ENCLOSURE DAMPERS.

HEATING CONTROL

THE HEATING WILL BE LOCKED OUT WHEN THE OUTSIDE AIR IS ABOVE THE HEATING LOCKOUT SET POINT (65 F ADJ.). WHEN ENABLED

DIRECT GAS FIRED HEATING

THE MICROPROCESSOR CONTROLLER WILL MODULATE THE DIRECT GAS BURNER TO MAINTAIN THE ACTIVE SUPPLY TEMPERATURE SETPOINT.

SUPPLY TEMPERATURE SET POINT CONTROL

THE ACTIVE SUPPLY TEMPERATURE SET POINT SHALL BE ADJUSTED REMOTELY BY THE BMS.

SUPPLY FAN OFF

FACTORY MOUNTED AND WIRED OUTDOOR AIR INLET DAMPER ACTUATOR IS DE-ENERGIZED AND SPRING RETURNS TO THE CLOSED POSITION.

SUPPLY FAN ON

FACTORY MOUNTED AND WIRED OUTDOOR AIR INLET DAMPER ACTUATOR IS ENERGIZED TO THE FULLY OPEN POSITION

SUPPLY AIR LOW LIMIT

IF THE SUPPLY AIR TEMPERATURE DROPS BELOW 35 F (ADJ.) FOR 300 SECONDS (ADJ.), THE CONTROLLER WILL DE-ENERGIZE THE UNIT AND GENERATE AN ALARM.

ALARM MANAGEMENT

THE MICROPROCESSOR CONTROLLER WILL MONITOR THE UNIT STATUS FOR ALARM CONDITIONS. UPON DETECTING AN ALARM, THE CONTROLLER WILL RECORD THE ALARM DESCRIPTION, TIME, DATE, AVAILABLE TEMPERATURES, AND UNIT STATUS FOR USER REVIEW. A DIGITAL OUTPUT IS RESERVED FOR REMOTE ALARM INDICATION.

ALARMS ARE ALSO COMMUNICATED TO THE BUILDING MANAGEMENT SYSTEM (BMS).

SUPPLY AIR DISCHARGE TEMPERATURE SENSOR ALARM

FAILURE OF THE SUPPLY AIR DISCHARGE TEMPERATURE SENSOR. UNIT IS SHUT DOWN.

SUPPLY AIR LOW LIMIT ALARM

SUPPLY AIR HAS FALLEN BELOW 35 F (ADJ.) FOR 300 SECONDS (ADJ.). UNIT IS SHUT DOWN.

DIRTY FILTER ALARM

INDICATES THE PRESSURE ACROSS THE FILTERS HAS INCREASED ABOVE THE DIRTY FILTER PRESSURE SWITCH SETTING (FIELDADJUSTABLE)

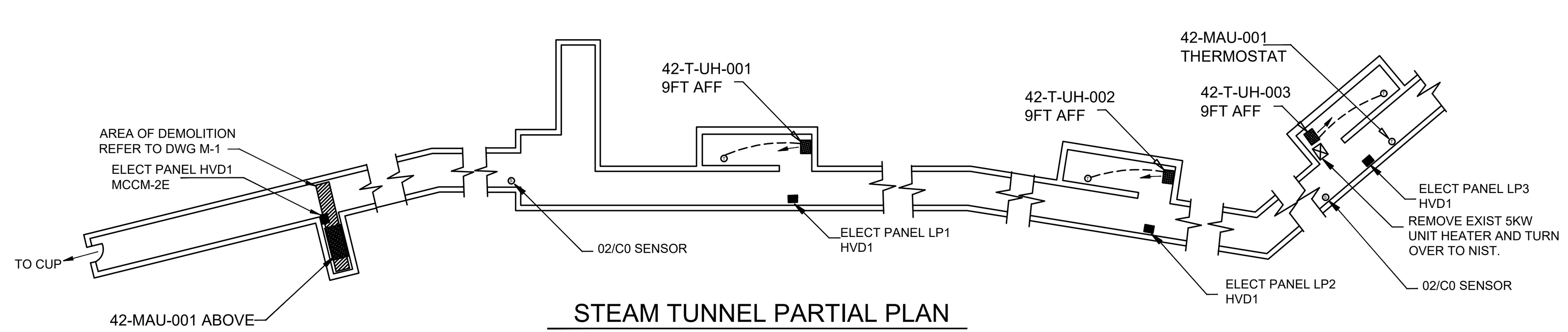
DIRECT GAS BURNER ALARM

INDICATES AN IGNITION CONTROLLER FLAME FAILURE ALARM. REQUIRES MANUAL RESET AT THE UNIT.

SUPPLY FAN ALARM

INDICATES THE SUPPLY FAN FAILED TO PROVE FOR A 30 SECOND (ADJ.) PERIOD.

MAKE UP AIR UNIT CONTROL SEQUENCE



STEAM TUNNEL PARTIAL PLAN

NOTES:
1. COORDINATE WITH NIST LOCATIONS OF ALL NEW SENSORS
2. REMOVE AND REPLACE 3 SPACE TEMP SENSORS, COORDINATE WITH NIST SENSORSS TO BE REPLACED.

REVISION	DESCRIPTION	BY	SUPV.	DATE
4				
3				
2				
1				

NATIONAL INSTITUTE OF STANDARDS AND TECHNOLOGY
BOULDER DESIGN AND CONSTRUCTION DIVISION
325 BROADWAY BOULDER, COLORADO 80305-3328

BUILDING 42 TUNNEL VENTILATION

DESIGN	DRAFTING	APPROVAL	DATE	SHEET NUMBER
COOK	COOK		3/10/2026	M-2

PROJECT NAME
ADDITIONAL TUNNEL HEAT

of